International and Monetary Economics

Course Code: 30054

Professor: Dmitriy Sergeyev

Lecture 12

Roadmap

1. Policy exercises

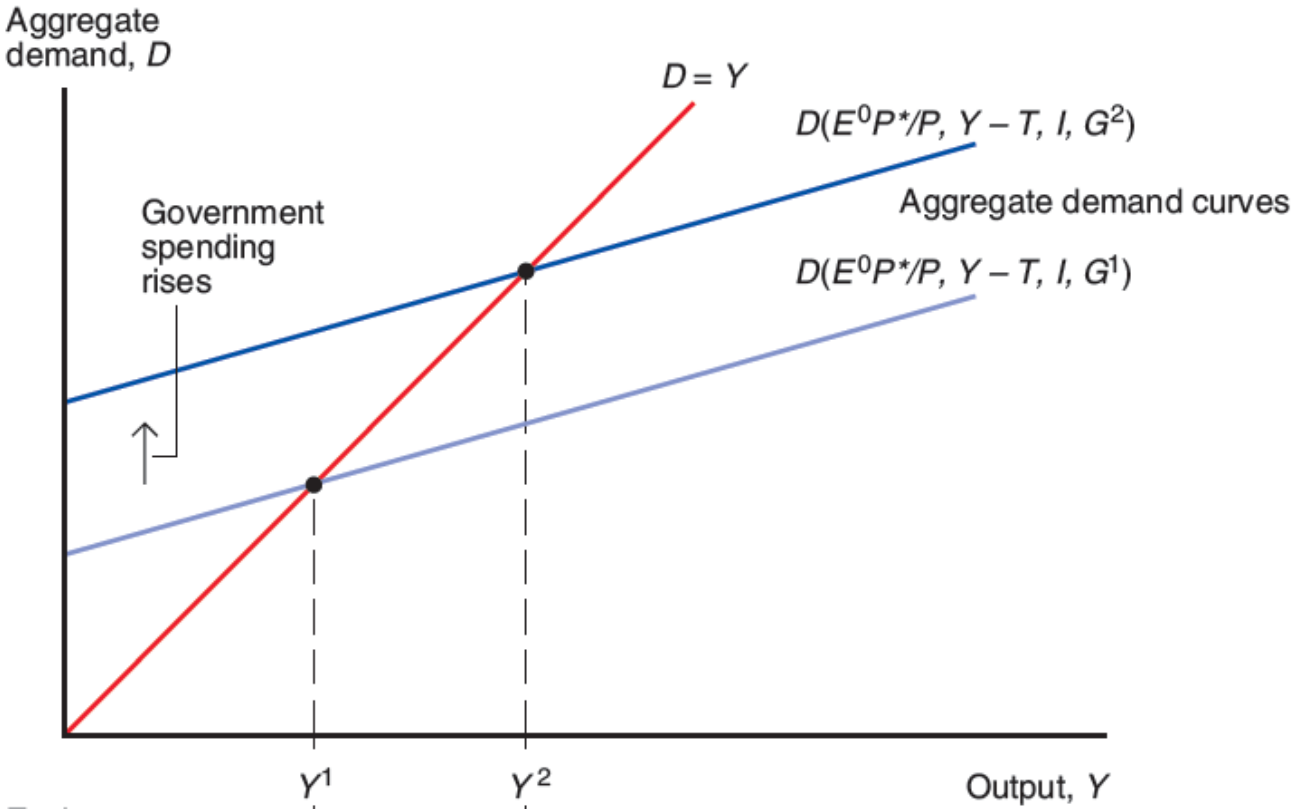
2. Liquidity trap

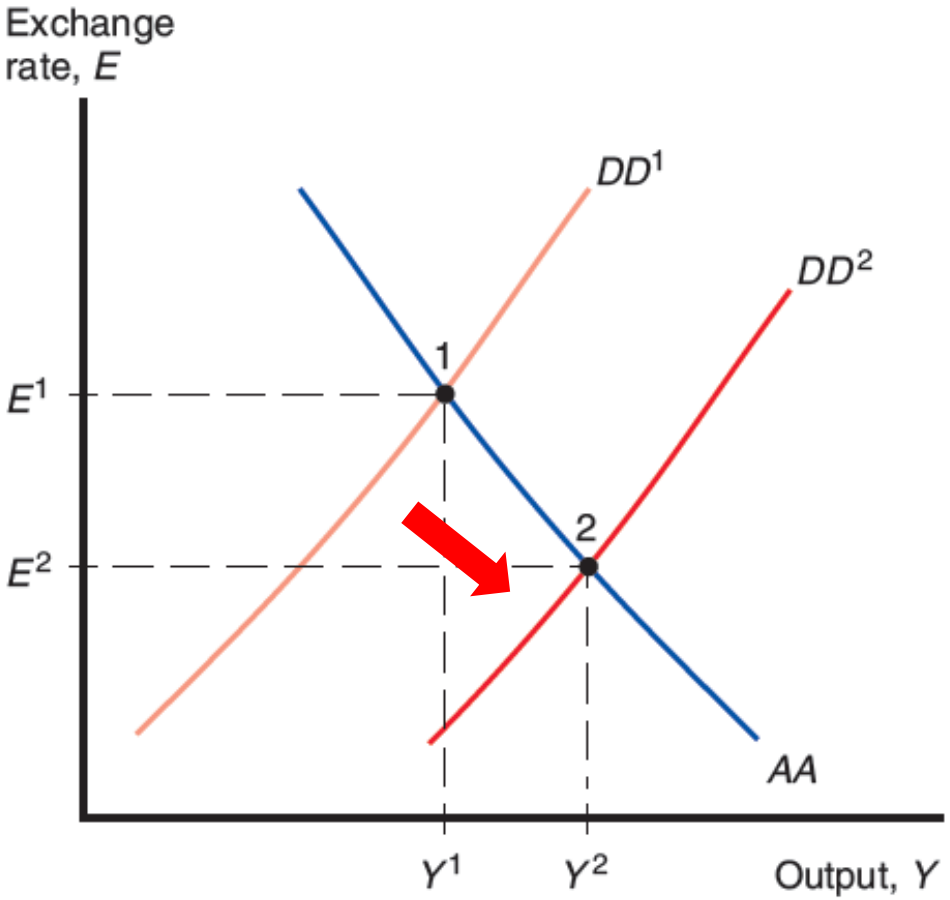
3. The XX curve

Policy Experiments

Exercise 2

A Temporary Change in Fiscal Policy





Exercise 2

Temporary Changes in Fiscal Policy

- An increase in government spending, a cut in taxes, or some combination of the two (i.e., expansionary fiscal policy) raises output

- The expansionary effect is partially offset by an appreciation of the currency

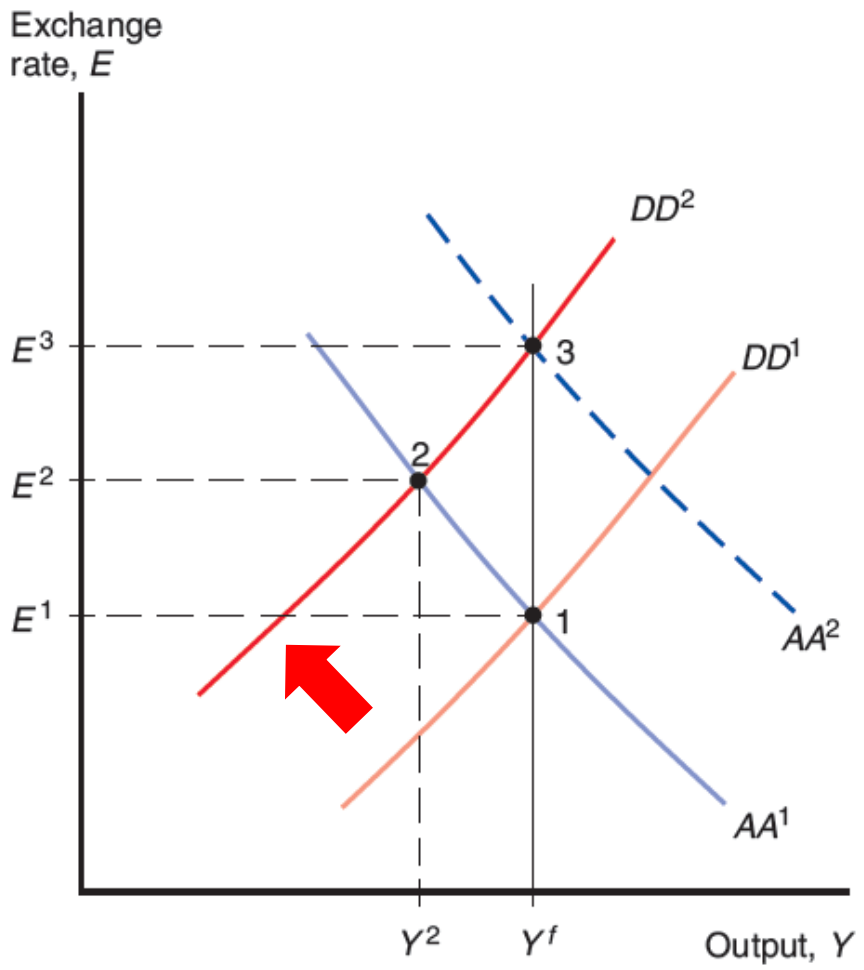
Business Cycles Stabilization

- Recessions
  - Temporary disturbances that lead to recessions can be offset through monetary or fiscal policies

- Booms
  - Temporary disturbances that lead to overemployment can be offset through contractionary monetary or fiscal policies

Example 3

A Temporary Fall in a World Demand for Domestic Products



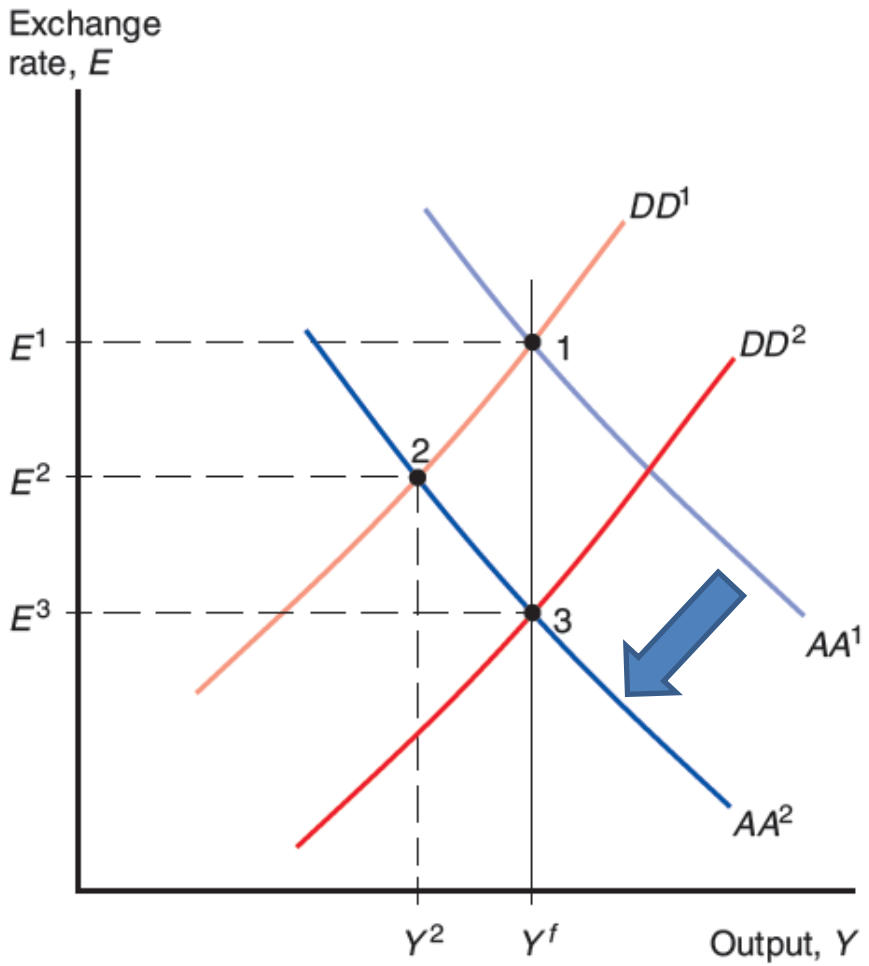
The shock moves the economy from point 1 to point 2.

A recession can be avoided by

1. a fiscal expansion (return to point 1)
2. a monetary expansion (point 3)

Example 4

A Temporary Money Demand Increase



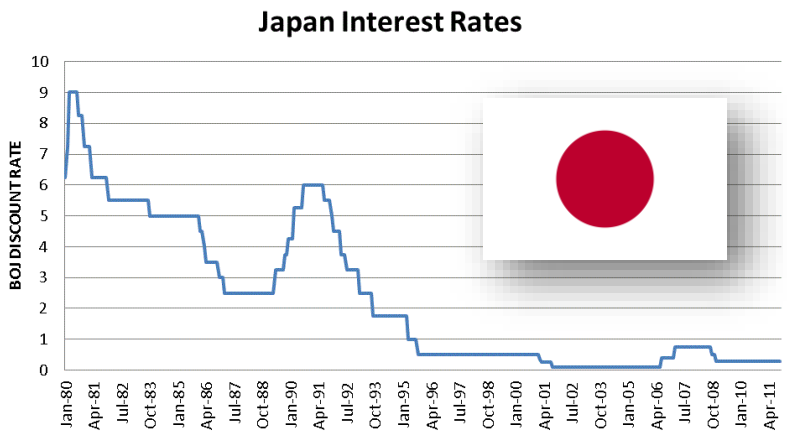
The shock moves the economy from point 1 to point 2.

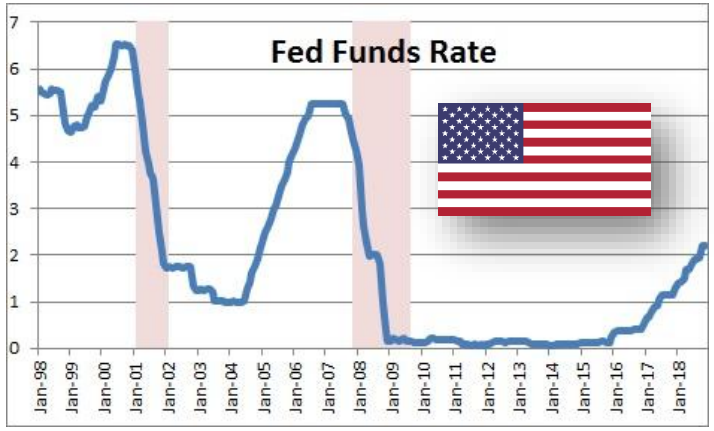
A recession can be averted by

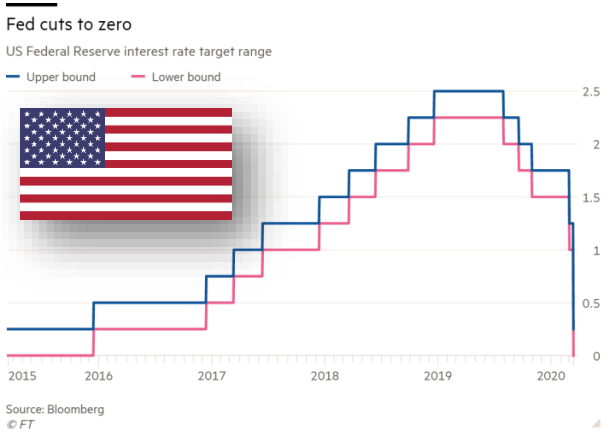
1. A fiscal expansion (point 3)
2. A monetary expansion (point 1)

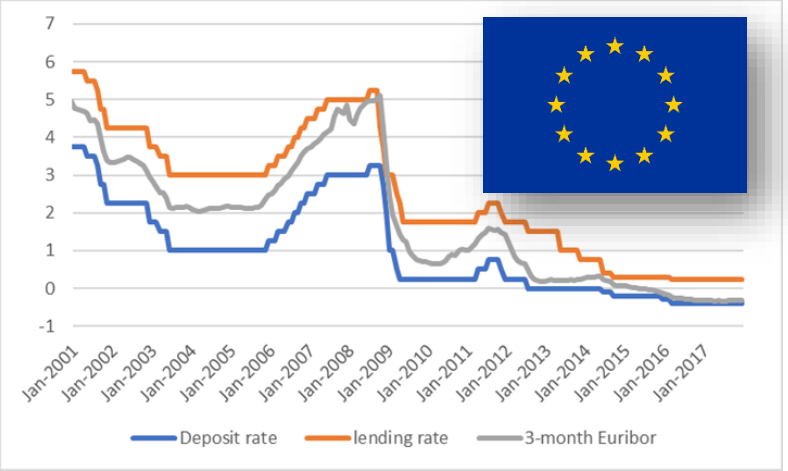
Liquidity Trap

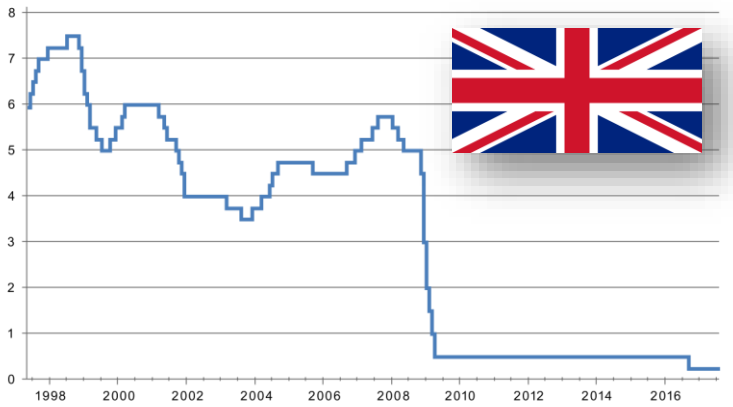
Liquidity Trap











Liquidity Trap

- When the nominal interest rate reaches 0 central bank cannot lower interest rates any further. Why?
  - In fact, central banks can reduce the interest rate slightly below zero but not too much

- Conventional monetary policy becomes ineffective

Liquidity Trap

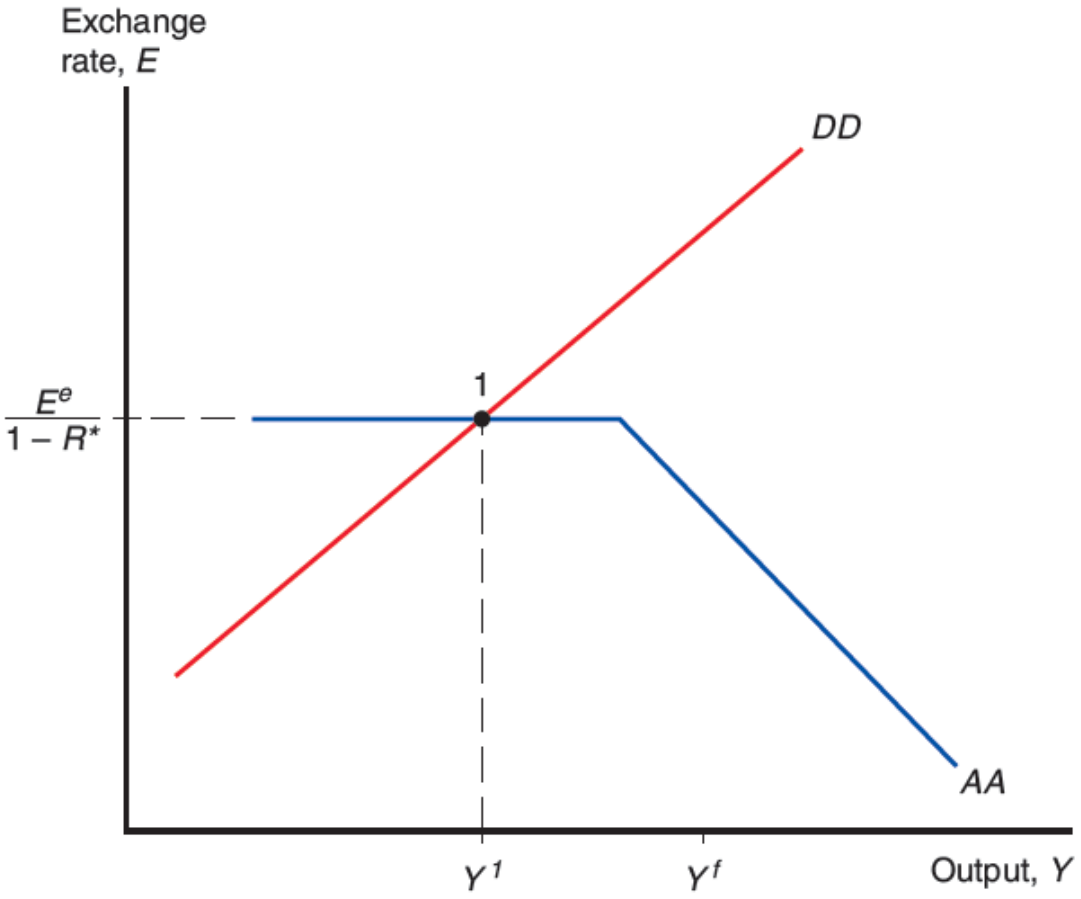
- The UIP

$$R = 0 = R^* + \frac{E^e - E}{E}$$

- Exchange rate does not depend on the current interest rate

$$E = \frac{E^e}{1 - R^*}$$

Conventional Monetary Policy Changes in Liquidity Trap



1. Temporary monetary expansion is **in**effective

   (shift in the AA curve to the right)

2. Permanent monetary expansion (**if believed**) can shift the AA schedule up!

Fiscal Policy Changes in Liquidity Trap

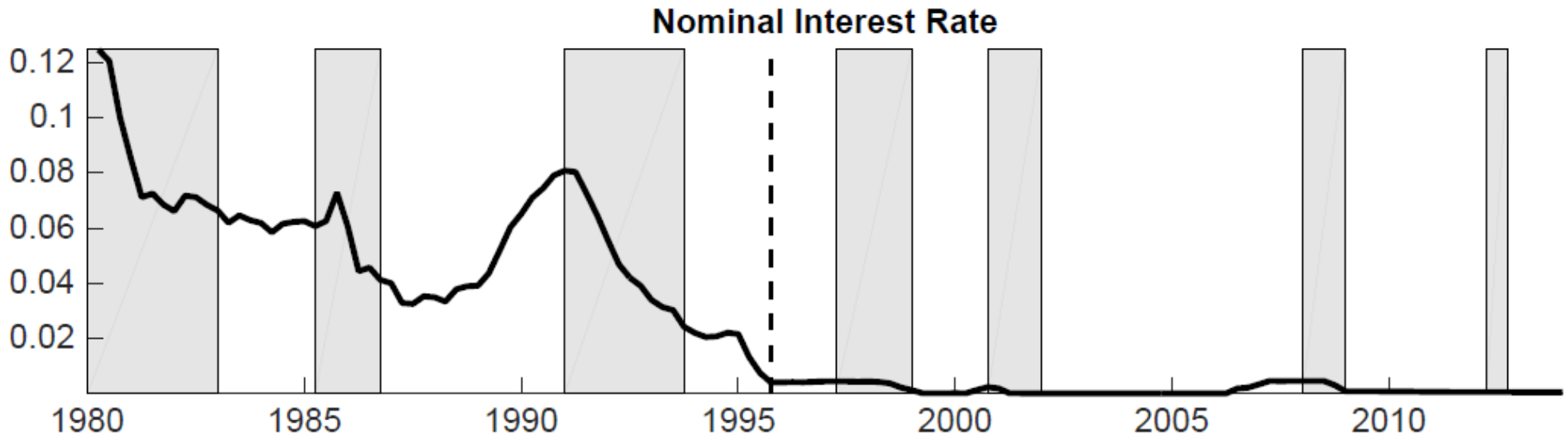
- **Outside of the liquidity trap**

  Fiscal expansion increases aggregate demand [present in the AA-DD model].

  But there are three opposing effects

  1. Future taxes increase (closed-economy macro) [absent from the AA-DD model]
  2. $R \uparrow \Longrightarrow C \downarrow, I \downarrow$ (closed-economy macro) [absent from the AA-DD model]
  3. Home currency appreciates (open-economy macro) [present in the AA-DD model]

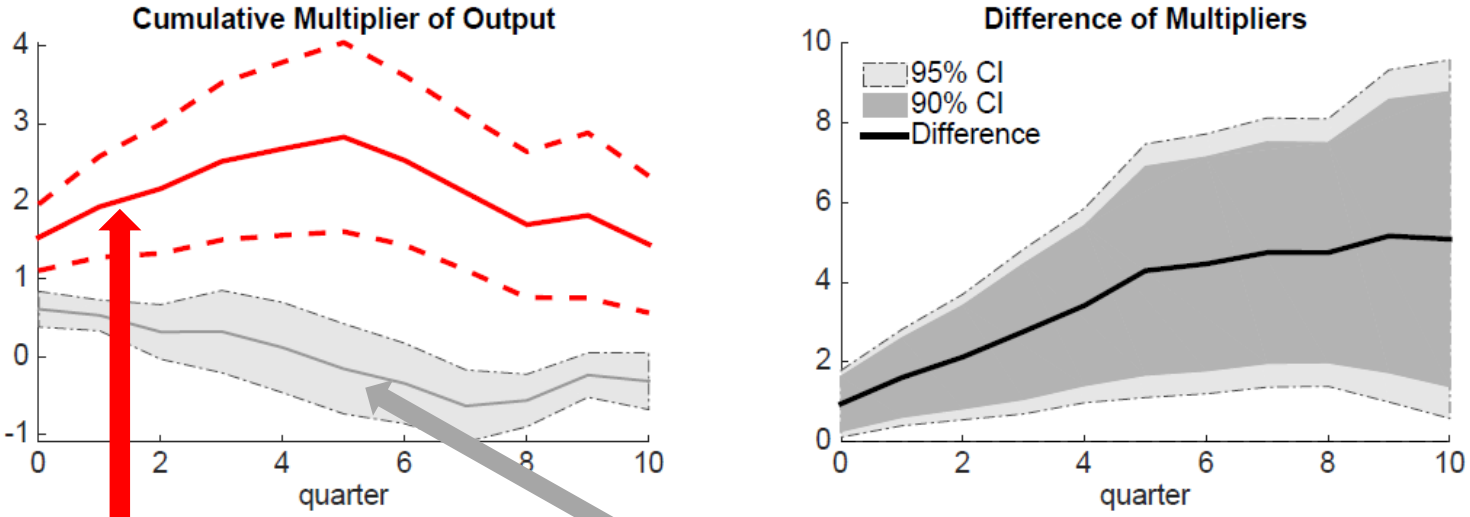
- **Liquidity trap**
  - The last two opposing effects are absent!
  - Do we see it in reality?



- **Japan has been in a liquidity trap since 1994! It is long enough to do a meaningful statistical analysis:**
  - Miyamoto, Wataru, Thuy Lan Nguyen, and Dmitriy Sergeyev. "Government spending multipliers under the zero lower bound: Evidence from Japan." *American Economic Journal: Macroeconomics* 10.3 (2018): 247-277.

$\frac{dY}{dG}$ in Japan

FIGURE 6. OUTPUT MULTIPLIERS AND THE DIFFERENCE IN THE MULTIPLIERS



Note: Output multipliers during normal and ZLB periods (left panel), and their difference (right panel). The borders around point estimates on the left panel are one-standard-deviation error bounds.

Source: Miyamoto, Nguyen, Sergeyev (2018)

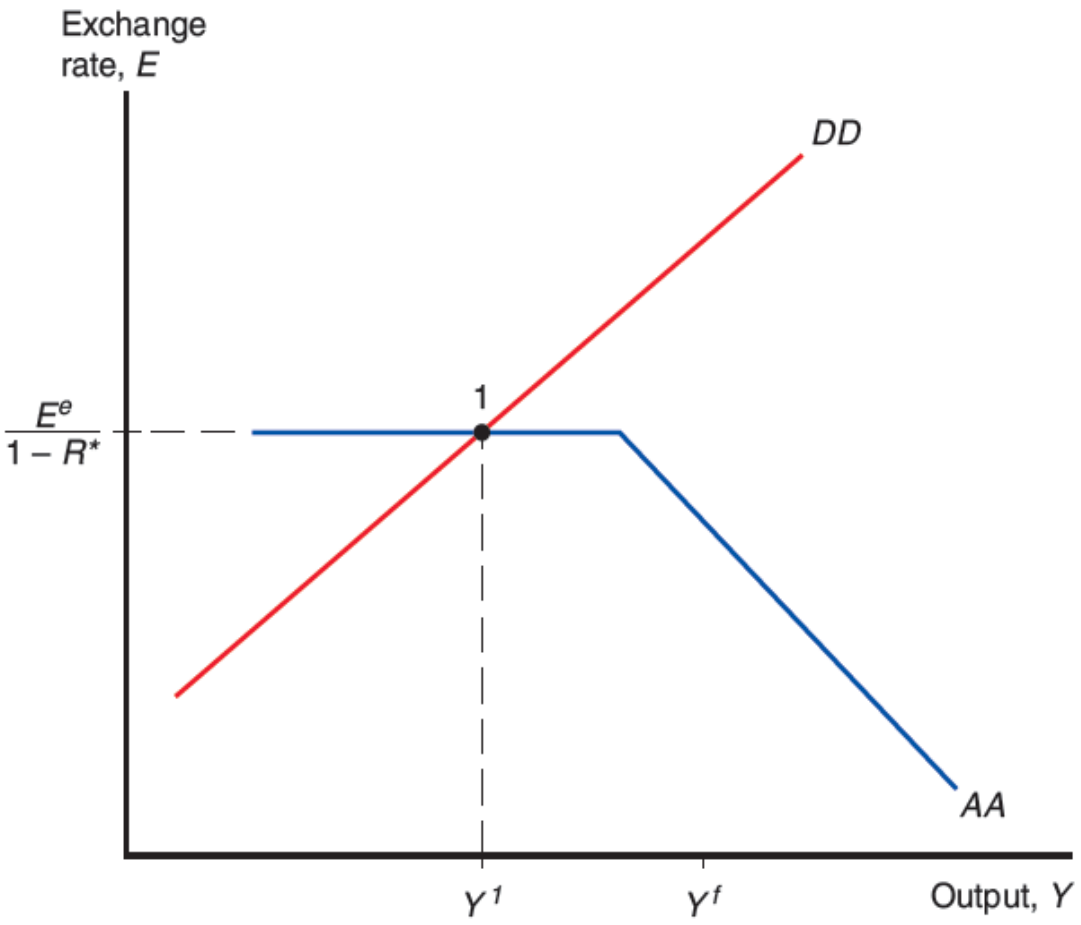
- The multiplier is higher in the liquidity trap than outside of it

Oil Shocks and the Economy

- **Conventional view on oil shocks**
  - Supply-side shocks affect aggregate supply [present in the AA-DD model]

- **Additional (aggregate demand) effects of oil shocks**
  - Poor households disproportionally suffer from spikes in gasoline prices [absent from the AA-DD model]
  - Inflation expectations increase with high oil prices [absent from the AA-DD model]
  - Oil importing status of a country [absent from the AA-DD model]
  - Monetary policy reacts to oil price shocks [present in the AA-DD model]

Oil Price Increase in the AA-DD Model



- Represent an oil price shock (an increase in oil prices) as identical increases in P and P* (because the oil market is global) and assume R* is unchanged (for simplicity)
- How does DD change?
- How does AA change?
- What happens to the output in
  - normal times?
  - in a liquidity trap?

Oil Shocks and Systematic Monetary Policy

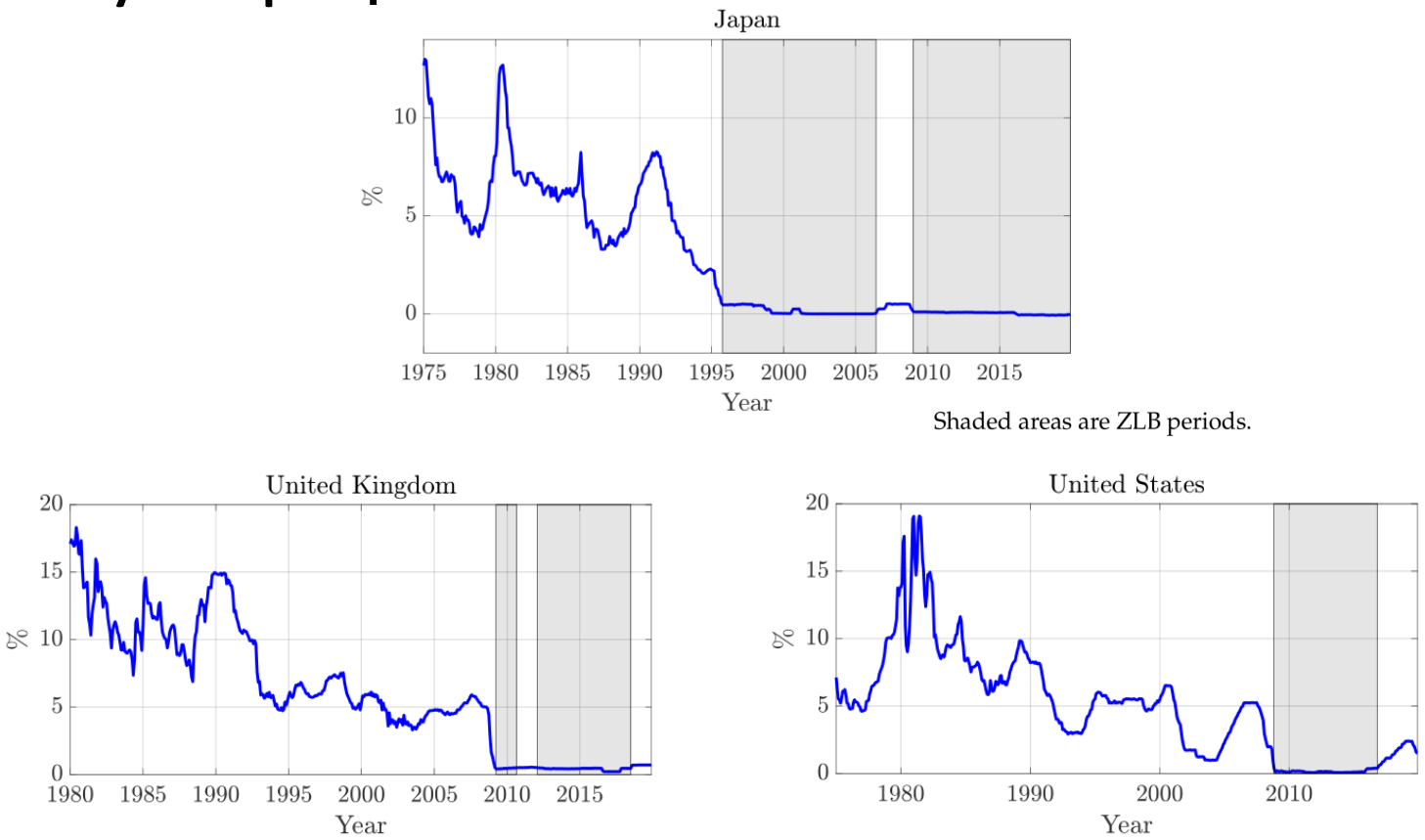
- **Previous slide**
  - Systematic reaction of monetary policy to shocks shapes the effects of oil (and other) shocks

- **Moreover**, this finding is true independent of the presence of other channels through which oil shocks propagate

- **Is it true in the data?**
  - Miyamoto, Wataru, Thuy Lan Nguyen, and Dmitriy Sergeyev. *How Oil Shocks Propagate: Evidence on the Monetary Policy Channel*. Centre for Economic Policy Research, 2024.

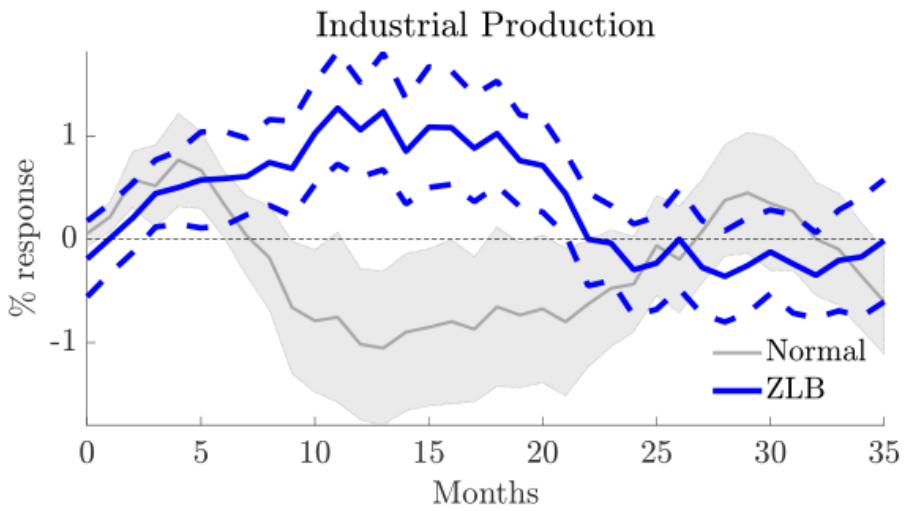
How to control for systematic monetary policy?

Study liquidity trap episodes



and be careful about the choice of oil shocks (see the paper)

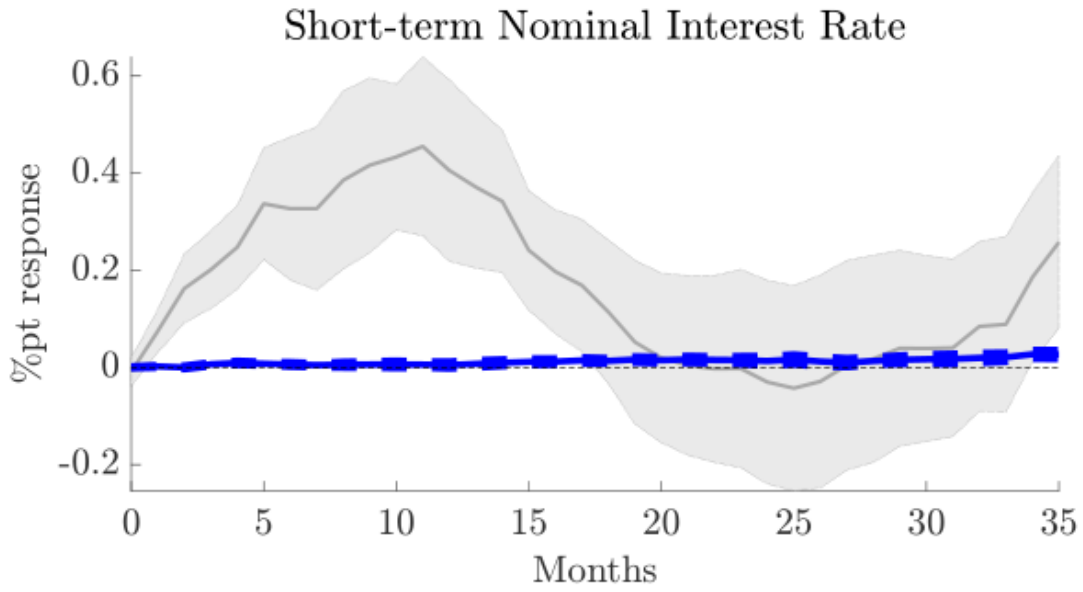
Japan

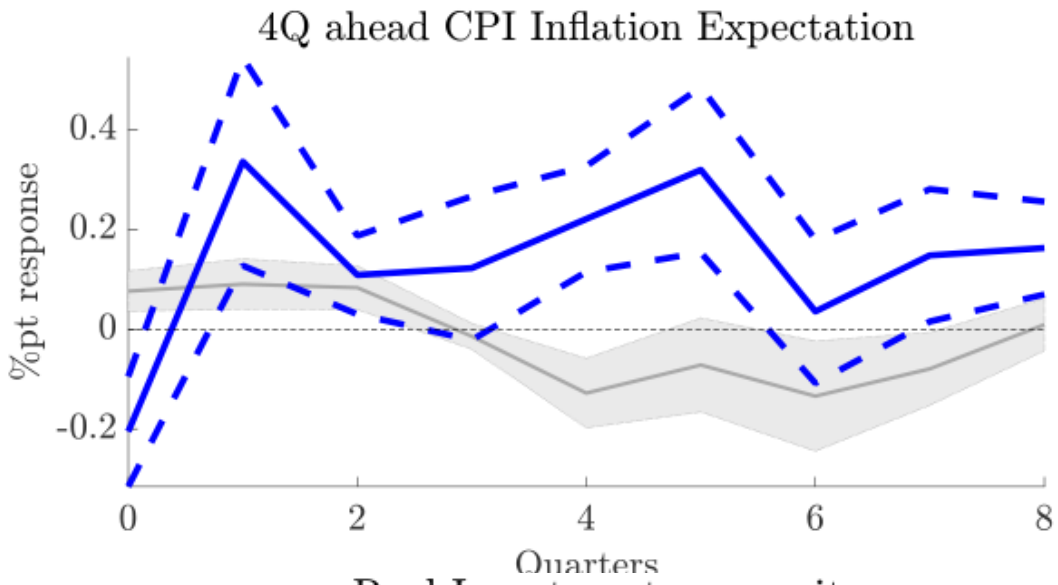


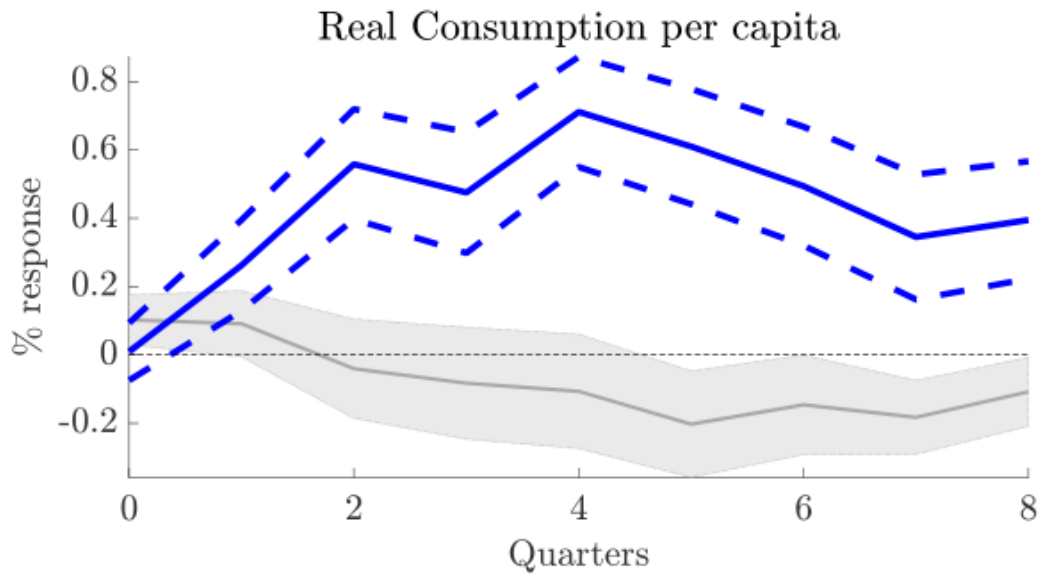
- Baseline specification comparing responses during **ZLB** and normal periods
- Oil shock is normalized to increase the oil price by 10% on impact in both periods
- Error bands at horizon h is $[\beta_h^y - \sigma_h^y; \beta_h^y + \sigma_h^y]$

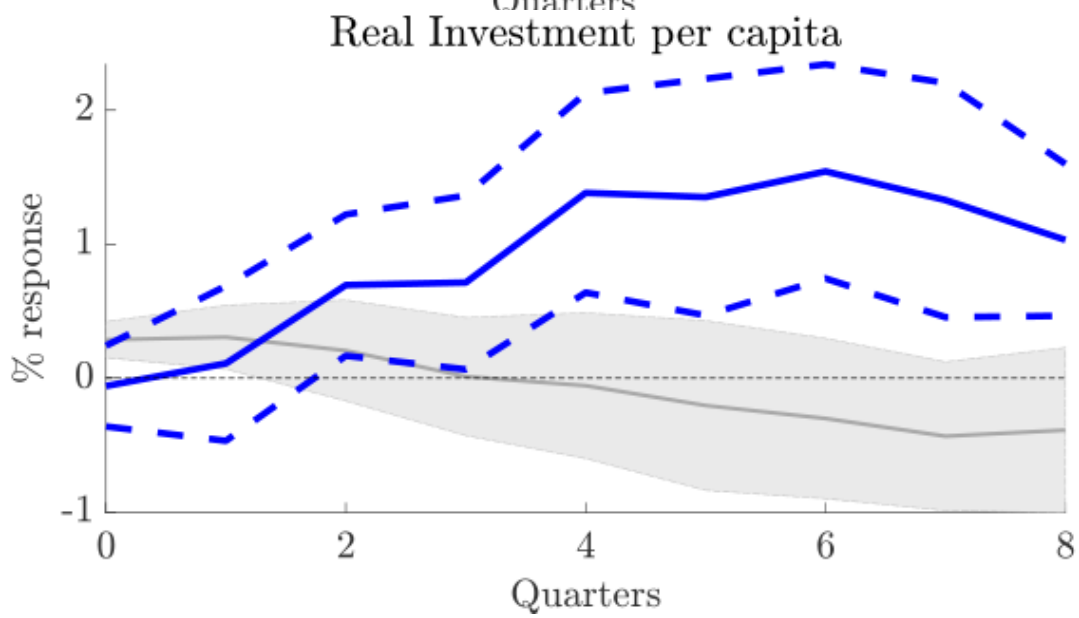
- Oil shocks seem to be expansionary in the liquidity trap in Japan. How come?
- Potential explanation: inflation expectations increase, reducing real interest rates and stimulating consumption and investment demand. Do we see this in the data?

The potential explanation seems to work in Japan

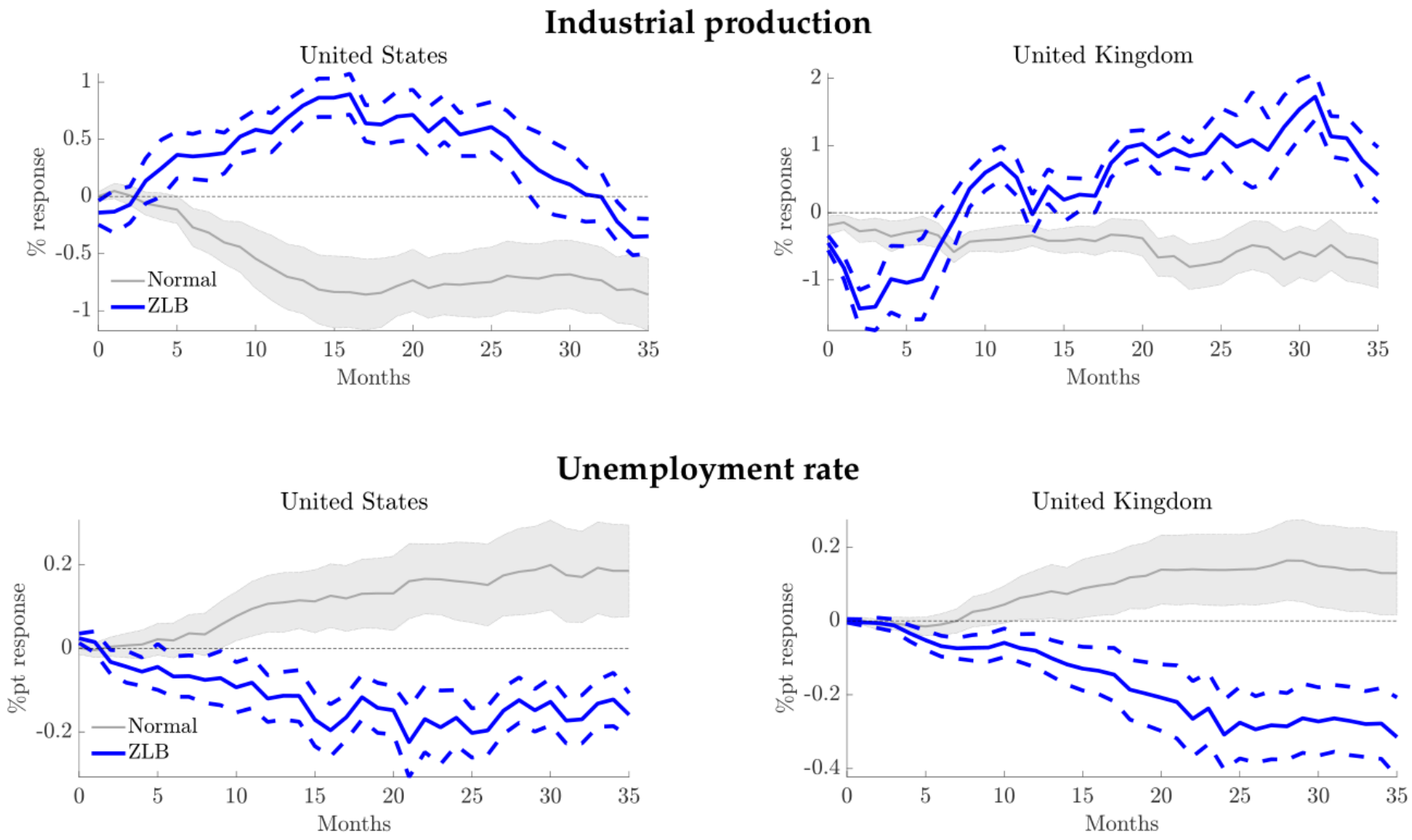








Evidence from other countries: US and UK



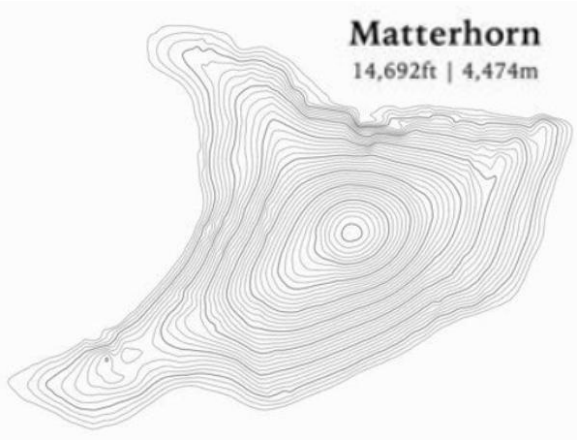
Policy and Trade Balance

How does NX change in our policy experiments?

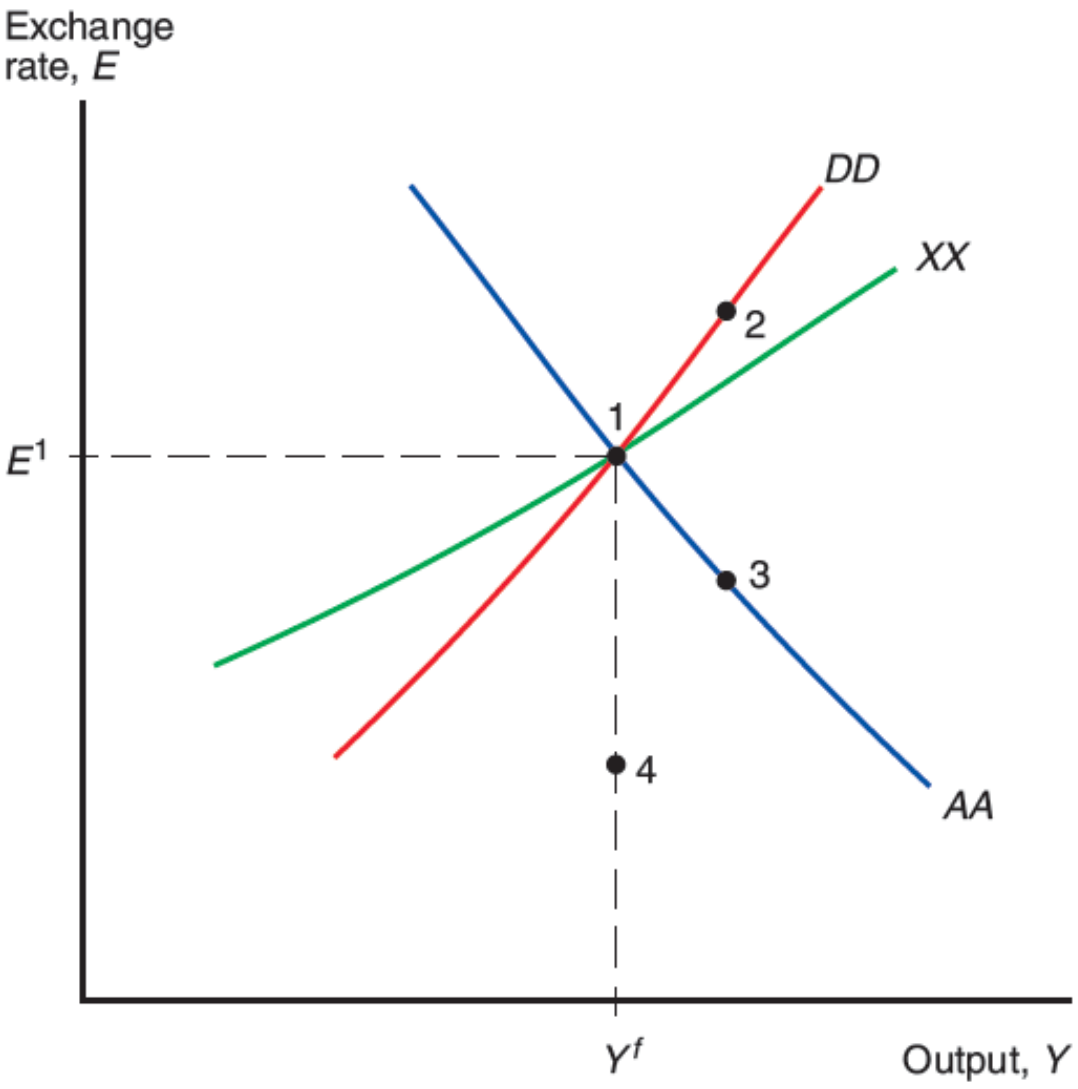
- Policymakers are often concerned about trade balance
  - Negative economic consequences
  - Political pressure
- A useful tool
  - The XX schedule shows combinations of the exchange rate and output at which the NX balance would be equal to some level X

$$NX\left(\frac{EP^*}{P}, Y - T\right) = X$$

  - This is like a topographical map that shows levels of constant elevation

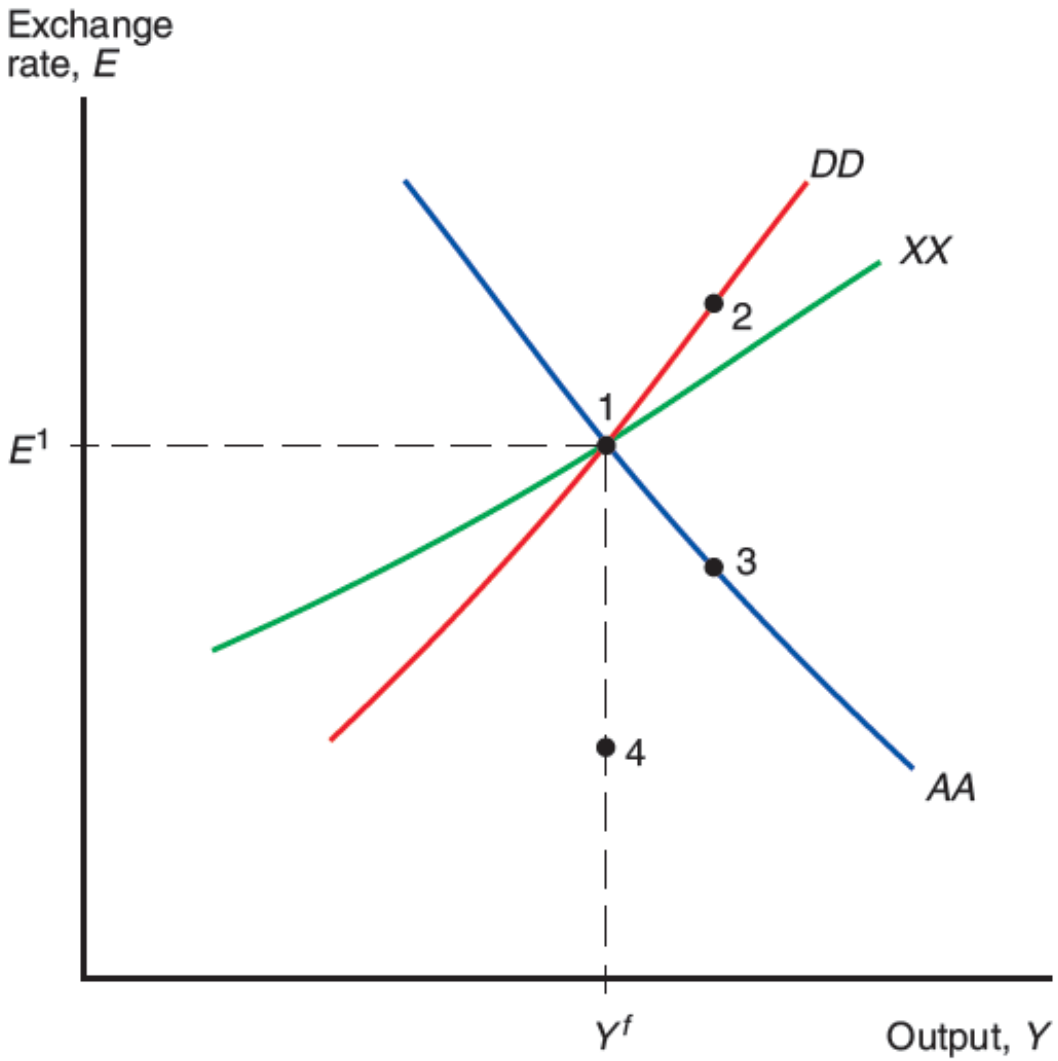


The XX Schedule



- Upward sloping
- Flatter than DD
  - it is easy to understand if you write the equation for the DD curve as Y-C(Y-T)-G-I=NX(EP*/P,Y-T). If Y goes up, then the left-hand side of this equation goes up, and hence, the right-hand side must go up. So, NX increases along the DD curve. As a result, the XX curve is flatter than the DD curve.
- $NX > X$ above the XX schedule, $NX < X$ below the XX schedule

The XX Schedule



- **Monetary expansion**
  - movement from 1 to 2
  - improves NX
- **Fiscal expansion**
  - movement from 1 to 3 (a temporary change)
  - reduces NX